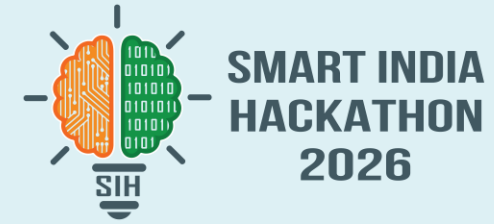
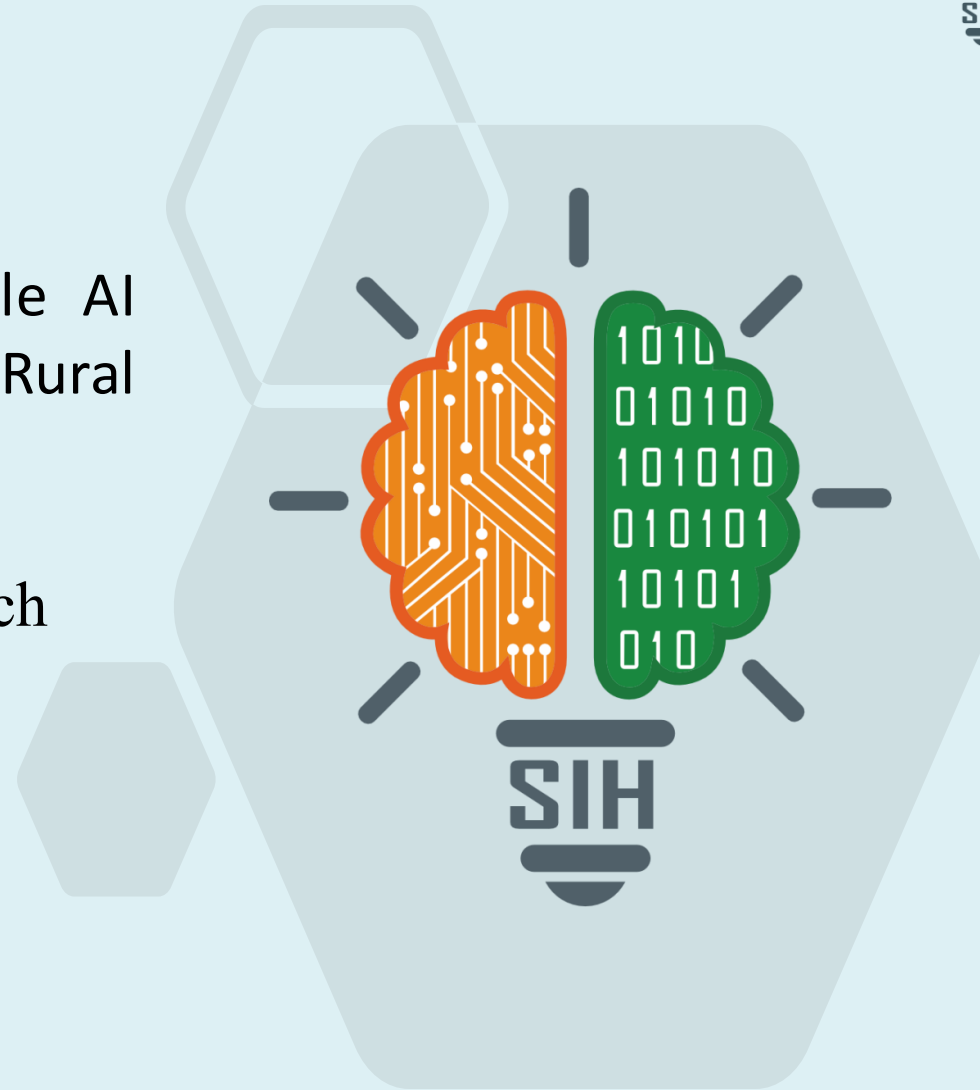


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26038
- **Problem Statement Title-** Explainable AI for Diabetic Retinopathy Screening in Rural India
- **Theme-** MedTech / Biotech / HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name-** Gryffindor



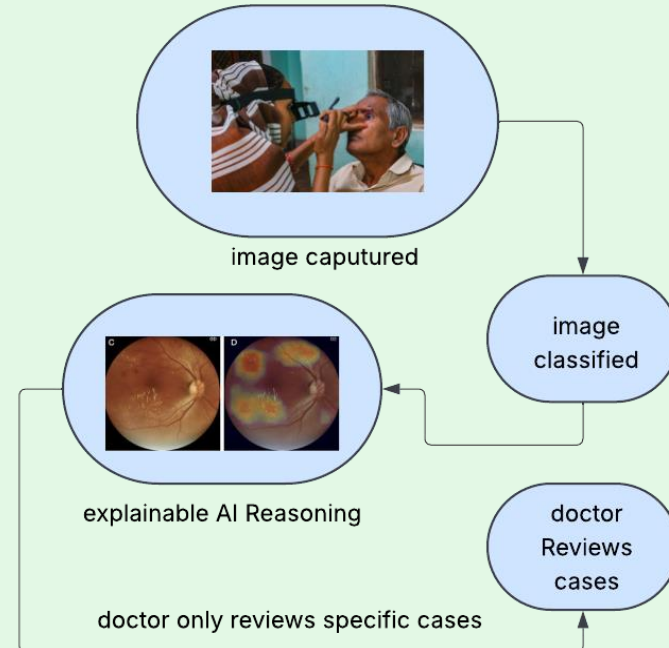
Proposed Solution

Solution overview

Core Idea: We are Building an AI-powered diabetic retinopathy screening system that analyses retinal fundus images and identifies the severity of diabetic retinopathy.

What are we aiming to provide:

1. Screens retinal images at rural PHCs.
2. Identifies normal and DR-affected eyes.
3. Provides an easy to understand screening result with visual evidence.
4. Flags patients who require specialist referral.



Key Innovation & Uniqueness

Informed Oversampling: Targeted or heuristic resampling to address class imbalance

Ensemble Model: Combine complementary models for robust and accurate DR prediction

Problem Fit

Rural Accessibility: Enables screening at local PHCs hence reducing the travel to specialised hospitals.

Early Diagnosis: Faster screening helps detect serious cases earlier.

Optimization of resources: Clearly flag patients who require further examination by an eye specialist, helping focus limited specialist resources where they are needed.

Technology Stack

MATLAB

Image Processing • Computer Vision • Deep Learning • Medical Imaging • Statistics & ML • Image Enhancement • Segmentation • Data Augmentation • Model Evaluation

AI

DR Grading • Explainable AI • Grad-CAM • Lesion Detection • CNNs • Transfer Learning • HITL • Sensitivity & Specificity

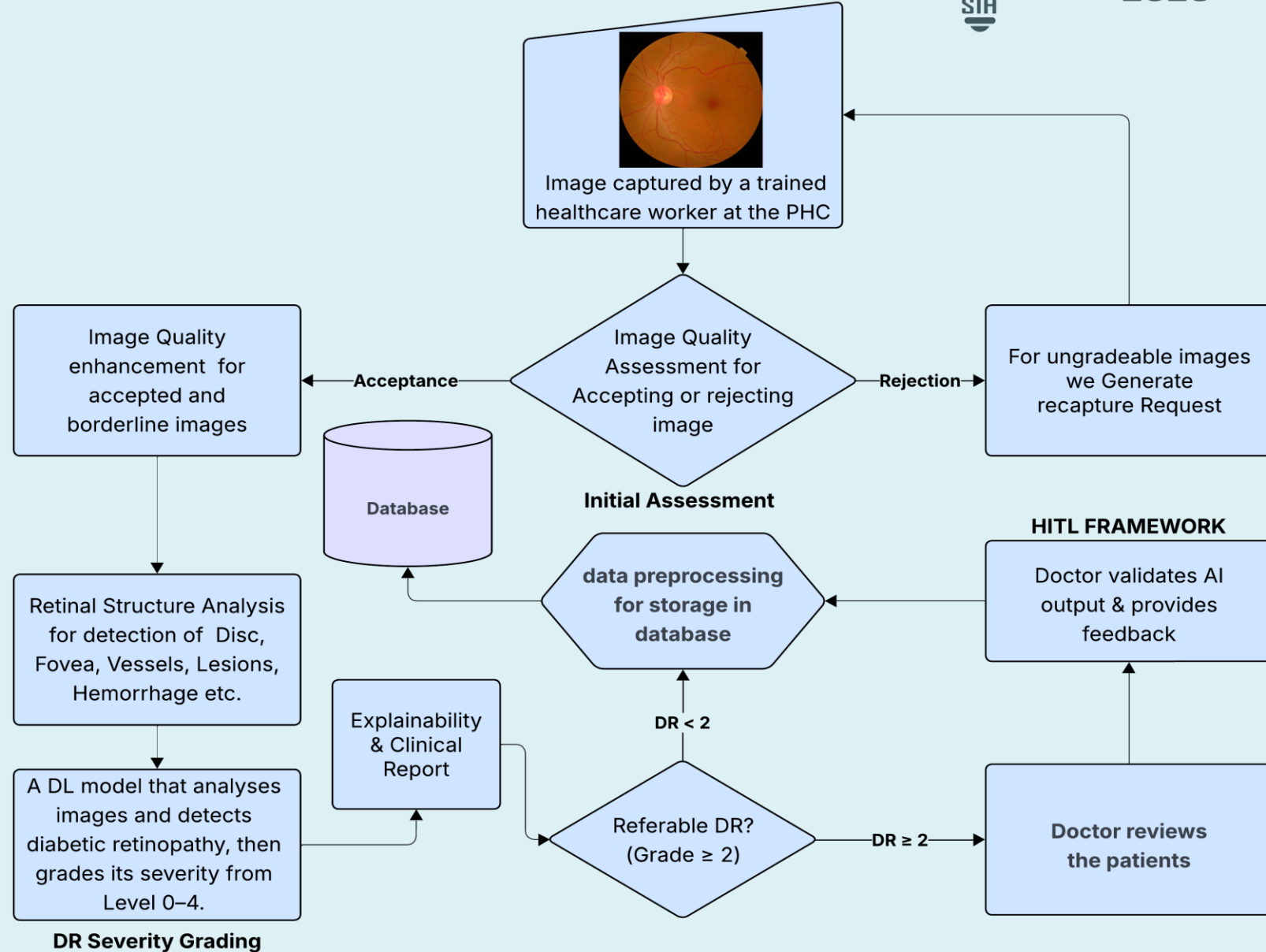
Simulation

Simulink • Bandwidth Modelling • Resource Allocation • Image Acquisition Rates • Throughput • Review Capacity

Web

MERN Stack • Vite • React • Tailwind CSS • Python/Fast API • REST APIs • MongoDB •

Link for website Prototype: [website](#)
Link for Github repository : [github](#)



Analysis of Deployment Challenges and Risks

1. Poor Rural Image Quality

Blur, glare & uneven lighting
Noisy images from cameras
Hard to know when an image is usable

2. Connectivity Problems

Unreliable rural internet
Large images are slow to upload
Delays in sending results

3. Specialist Shortage

Too few ophthalmologists
High workload from routine screening
Difficult to prioritise urgent cases

4. Privacy and Integration

Patient data needs secure handling
Difficult integration with existing systems
Limited transparency in AI decisions

Problem - Low number
of Ophthalmologist in
India

Our approach to
solve these problem

Solution –
Optimize the
allocation of
resources we have

Strategies for Mitigation & Engineering Viability

1. Better Image Quality

Check image quality before screening
Improve blur, lighting & noise
Prompt recapture when needed

2. Works with Poor Connectivity

Compress images for faster transfer
Support low-bandwidth environments
Process images locally when possible

3. Reduce Specialist Workload

Automatically screen retinal images
Highlight cases needing attention
Prioritise patients with high risk

4. Secure & Connected

Protect patient data during transfer
Generate simple screening reports
Support integration with existing systems

IMPACT AND BENEFITS

PRIMARY IMPACT

Earlier Detection

Identify diabetic retinopathy at an early stage.

Faster Referrals

Prioritize patients who need specialist attention.

Rural Accessibility

Make screening accessible at rural health centres .

SECONDARY IMPACT

Reduced Workload

Reduce routine screening burden on ophthalmologists.

Explainable Decisions

Help healthcare workers understand AI predictions.

Scalable Screening

Support wider diabetic eye screening across rural areas.

BENEFITS OF THIS WORK

Better Vision Outcomes

Early detection can help prevent avoidable vision loss.

Faster Treatment

High risk patients can reach specialists sooner.

Reduced Specialist Burden

Doctors can focus more on patients who need attention.

Cost Savings

Reduce unnecessary travel and specialist consultations.

Greater Trust in AI

Trust of healthcare worker in AI will increase

[Gulshan, V., Peng, L., Coram, M., et al. \(2016\).](#) Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA*, 316(22), 2402–2410.

[Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. \(2017\).](#) Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 618–626.

[Natarajan, S., Jain, A., Krishnan, R., Rogye, A., & Sivaprasad, S. \(2019\).](#) Diagnostic Accuracy of Community-Based Diabetic Retinopathy Screening With an Offline Artificial Intelligence System on a Smartphone. *JAMA Ophthalmology*, 137(10), 1182–1188.

DATASET

APTOS 2019 — DR severity classification (Grade 0–4).

IDRiD — DR grading and lesion-level annotations (microaneurysms, hemorrhages, exudates)

Messidor-2 — External validation of DR screening models

DRIVE — Retinal vessel segmentation

[Porwal, P., Pachade, S., Kamble, R., Kokare, M., Deshmukh, G., Sahasrabudhe, V., & Meriaudeau, F. \(2018\).](#) Indian Diabetic Retinopathy Image Dataset (IDRiD): A Database for Diabetic Retinopathy Screening Research. *Data*, 3(3), 25.

[Wong, T. Y., Sun, J., Kawasaki, R., et al. \(2018\).](#) Guidelines on Diabetic Eye Care: The International Council of Ophthalmology Recommendations for Screening, Follow-up, Referral, and Treatment Based on Resource Settings. *Ophthalmology*, 125(10), 1608–1622.

[Rajalakshmi, R., Subashini, R., Anjana, R. M., & Mohan, V. \(2018\).](#) Automated diabetic retinopathy detection in smartphone-based fundus photography using artificial intelligence. *Eye*, 32, 1138–1144.



“The concept that this is a microvascular disease is incomplete.”
— Thomas W. Gardner,